

Yu-Sen (Eason) Wu

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Education

Northwestern University, Evanston, IL

MS in Machine Learning & Data Science | Sep 2025 – Present

National Tsing Hua University (NTHU), Hsinchu, Taiwan

BS & MS in Industrial Engineering & Engineering Management | Sep 2019 – Jun 2024

Technical Skills

- Programming & Tools: Python (Pandas, NumPy, PyTorch, TensorFlow), SQL, R, Java, Dart, JavaScript, Git, Docker
- Data & Cloud: Data Modeling, ETL, Data Warehousing, Pipeline Architecture, Spark, Apache, AWS, GCP, CI/CD
- Analytics & ML: Data Visualization (Tableau, Power BI, Matplotlib, Seaborn), Machine Learning, NLP, Time Series Analysis, A/B Testing, Feature Engineering, Statistical Modeling, Generative AI, AI agent, agentic RAG
- Domain Knowledge: KYC/AML/CFT Compliance, Risk Analytics

Work Experience

KYC Analyst Intern | WOOX | Jun 2025 – Present

- Built a robust JavaScript/ETL and CI/CD pipeline to automate compliance content tracking across 19 crypto exchanges, reducing manual review workload from 10 hrs/week to less than 10 minutes.
- Built an automated tool using Google Apps Script and LLM to generate dynamic tables of country restrictions and AML summary report emails, eliminating manual updates and cutting review time by 90%.
- Built an RAG AI-agent (LangChain, Python, prompt engineering) integrated with PostgreSQL and APIs that interprets questions, self-generates, and executes SQL, cutting query time by 95% and ensuring 100% auditability.

Gait Recognition & Safety AI Researcher | NTHU | Jul 2022 – Dec 2024

- Built and deployed a deep learning solution on head-worn IMU data streams to detect slip patterns 0.1 s before balance loss; integrated haptic alerts and achieved 96% prediction accuracy in a field pilot with construction workers.
- Applied signal processing feature engineering, improving identification accuracy by 31% over baseline.

Selected Projects

TutorMatching Platform (AI-Powered EdTech)

- Developed an AI-powered education platform combining instant problem-solving with professional tutor matching, supporting photo-based question uploads via Next.js, TypeScript, and Firebase; reached 100+ users in the first week.
- Designed an AI-driven solution that achieved 98%+ accuracy in solving math problems and reduced operational costs by 70%, ensuring equitable access for students from diverse income backgrounds.

Mapify – Food Map Social App

- Developed a Flutter-based app that parsed Instagram posts by integrating LLM to auto-geocode Instagram data.
- Optimized data-entry workflow by reducing manual input per spot from 3 minutes to under 20 seconds through automation; increased daily active pins by 5x, significantly improving user activity and retention.

Publication

"Utilization of two types of feature datasets with image-based and time series deep learning models in recognizing walking status and revealing personal identification." Advanced Engineering Informatics. (SCI, Published, 2024)

- Proposed a novel approach for gait pattern recognition and personal identification using image-based and time-series deep learning models, achieving an accuracy of up to 99% in various walking scenarios.